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A Highly Conducting Polymer for Self-Healable, Printable, and Stretchable Organic Electrochemical Transistor Arrays and Near Hysteresis-Free Soft Tactile Sensors

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A stretchable and self-healable conductive material with high conductivity is critical to high-performance wearable electronics and integrated devices for applications where large mechanical deformation is involved. While there has been great progress in developing stretchable and self-healable conducting materials, it remains challenging to concurrently maintain and recover such functionalities before and after healing. Here, a highly stretchable and autonomic self-healable conducting film consisting of a conducting polymer (poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate), PEDOT:PSS) and a soft-polymer (poly(2-acrylamido-2-methyl-1-propanesulfonic acid), PAAMPSA) is reported. The optimal film exhibits outstanding stretchability as high as 630% and high electrical conductivity of 320 S cm^{-1}, while possessing the ability to repair both mechanical and electrical breakdowns when undergoing severe damage at ambient conditions. This polymer composite film is further utilized in a tactile sensor, which exhibits good pressure sensitivity of 164.5 kPa^{-1}, near hysteresis-free, an ultrafast response time of 19 ms, and excellent endurance over 1500 consecutive presses. Additionally, an integrated 5 × 4 stretchable and self-healable organic electrochemical transistor (OECT) array with great device performance is successfully demonstrated. The developed stretchable and autonomic self-healable conducting film significantly increases the practicality and shelf life of wearable electronics, which in turn, reduces maintenance costs and build-up of electronic waste.

1. Introduction

Conjugated polymers with features of flexibility, stretchability, and printability have huge potential applications in soft robotics, health monitoring, and human-machine interfaces.[1–5] One challenge in the development of these soft electronic devices is the requirement for long-term durability.[6] The mechanical breakdown due to scratching, delamination, bending, and stretching seriously limits their practicality and lifetime. The reduced durability results in increased maintenance costs and electronic waste build-up. The self-healing ability of materials, for example, automatic closure of cracks, can significantly improve the reliability and lifetime of electronic devices.

Over the last decades, introduction of more flexible molecular building blocks[1,7] has been pursued to develop intrinsically stretchable and self-healable (semi) conducting polymers. Another vital and effective strategy is the incorporation of soft-material matrix into the conducting polymer, where the soft material provides self-healing capability and enhances the stretchability of the active conducting material.[8–13] Among the various intrinsic conducting polymers, the current benchmark, poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS)—a blend polymer with the highest reported intrinsic conductivity, shows the greatest prospective to develop high performing self-healable conductors using the soft-material/conducting polymer blend strategy.[14–16] Electrical healing of PEDOT:PSS-based system has been demonstrated by exposure to water or water vapor. Here, the water uptake allowed the swelling of PSS segment, thereby filling the damaged area.[17] While, due to the existing Coulombic interaction between PEDOT and PSS phases, the charge transport path is repaired along with the restoration of the PSS phase, and thus its electrical conductivity.[17] Self-healable PEDOT:PSS system is also realized by introducing a variety of additives such as Triton X-100,[18–20] 4-dodecylbenzenesulfonic acid,[21] Polyethylene glycol,[22] etc., by altering the wetting and viscoelastic properties of the PEDOT:PSS composites. The healing mechanism heavily relies on their water-uptake capability, leading to the reformation of hydrogen bonds and thus repairing the damaged film with water stimulus or external pressure.[17,18,20–22] Nonetheless, most of these healable PEDOT:PSS films exhibit relatively poor mechanical properties,[22,23] and their mechanical properties are further compromised after healing.[17,20,22] At the same time, a number

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of efforts have also been put in to improve the mechanical properties of PEDOT:PSS. For instance, incorporation of soft-polymer matrix, such as poly(vinyl alcohol),[12,24] polyurethane,[8] polyacrylic acid,[9] have been attempted to boost the mechanical property of PEDOT:PSS. The weak molecular interactions including hydrogen bonds and/or ionic bonds between the conducting polymer and soft materials can be easily broken to allow energy dissipation upon strain, making the system more tolerant of strain and mechanical stimuli. However, the enhanced mechanical property of these systems can often compromise their electrical properties. It is thus no easy feat to achieve and maintain high conductivity, high stretchability, and self-healability simultaneously for conducting polymers.[17] Another challenge is that when the damaged area is comparable to the thickness of film, it is difficult to realize an efficient self-healing process.[25] Therefore the self-healing process is currently limited to only small-area damages or thick polymer layers (>1–100 um) are considered. However, ideally, a self-healable conducting thin film is desirable for developing flexible wearable electronics (like e-skin).

In this report, we successfully demonstrate, for the first time, a highly stretchable and autonomic self-healable conducting polymer with great electrical conductivity, via incorporating dynamic intermolecular bonds, including hydrogen bonds and ionic bonds between the conducting polymer of PEDOT:PSS and a soft-material of poly(2-acrylamido-2-methyl-1-propanesulfonic acid (PAAMPSA) (**Figure 1**a). The hydrogen and ionic bonds formed between PEDOT:PSS and PAAMPSA are able to undergo an energy dissipation mechanism through breakage of bonds when strain is applied or when severe damage occurs from mechanical stretching or blade cutting, while these dynamic bonds can be reformed to repair the damaged area and thus restoring its mechanical property at ambient conditions (≈23 °C RT, ≈50% RH). An ionic liquid (IL) of 1-Ethyl-3-methylimidazolium – trifluoromethanesulfonate (EMIM OTF, Figure 1a) is incorporated to induce the higher PEDOT crystallinity facilitating the charge transport and thus strengthening the electrical conductivity of the composite. The optimal PEDOT:PSS/PAAMPSA/IL film exhibits stretchability as high as 630% and high electrical conductivity of 320 S cm^{-1}, while possessing the ability to repair both mechanical and electrical breakdowns when undergoing severe damage at ambient conditions. The film exhibits high stretchability after severe damage-and-heal, with a maximum strain of 510%. Notably, no significant change in Young's modulus (≈0.8 MPa) was observed before and after healing. To examine the versatility of this polymer composite, we incorporated it as functional components in tactile sensors as well as in organic electrochemical transistors (OECTs). We demonstrated a tactile pressure sensor exhibiting high-pressure sensitivity (164.5 kPa^{-1}), ultrafast response time as low as 19 ms, and superior operational stability with negligible performance change after 1500 cycling test. The fabricated tactile sensor presents great practicality to be used in human-machine interaction systems to identify the weight of grabbed objects. Additionally, a large-area, stretchable OECT array with high device performance (in terms of the high transconductance of 12.95 ± 1.85 mS, fast transient speed of 7.1 ms, and good electrochemical doping/dedoping stability) was presented. This OECT array, having a thin channel thickness of ≈200 nm, exhibits remarkable self-healability at ambient conditions, which further indicates its utility for e-skin electronics. Our findings provide a facile, low-cost but effective strategy to develop a highly stretchable and autonomic healable conductor towards deformable electronics and future large-area industrial production.

2. Results and Discussion

Among the various dynamic bonds, non-covalent interactions such as hydrogen and ionic bonds are particularly suitable for skin-inspired electronics owing to their spontaneous formation and healing ability. A soft-polymer of PAAMPSA, the molecular structure of which is depicted in Figure 1a, contains amide and sulfate groups with moderate hydrogen-bonding strength, which allows the formation of a polymer network with PEDOT:PSS (Figure 1b) and enables good miscibility as well as decreases the tensile modulus of the nanocomposite. PAAMPSA also serves as the regulator of water content in the nanocomposite benefitting from its highly hygroscopic nature at room temperature, which is crucial for the self-healing process.[17,20,22] Considering these desirable features, PAAMPSA was used as the soft material of choice to reinforce the mechanical property and introduce dynamic bonds between the conducting polymer chains, to improve the self-healing capability. Initial evaluation of the electrical conductivity (σ) was performed using a four-point method. Pristine PEDOT:PSS shows an electrical conductivity of 1.1 S cm^{-1} (Table S1, Supporting Information). No significant change was found in the σ value when introducing 67% of PAAMPSA in PEDOT:PSS (0.3 S cm^{-1}), but its mechanical property was significantly enhanced with a maximum strain of 390% (Figure 1c). Our previous work found that incorporation of water-soluble ionic liquid of EMIM OTF (Figure 1a) with strongly electron-withdrawing CF_3 moieties into PEDOT:PSS triggered molecular rearrangement, resulting in enhanced packing of PEDOT chains and larger PEDOT-rich domains, thus facilitating charge transport and significantly improving its electrical conductivity.[23,26] It was also found that PEDOT:PSS films loaded with 50 wt% of EMIM OTF had the highest conductivity value of 1018 S cm^{-1} (Figure S1, Supporting Information). Further increasing the EMIM OTF content led to solution precipitation affecting the film processing, therefore, the optimal loading of EMIM OTF in the nanocomposite used in this report is equal to the mass of PEDOT:PSS, unless otherwise specified. Incorporation of EMIM OTF into the PEDOT:PSS/PAAMPSA nanocomposite also brings a remarkable enhancement in its electrical conductivity, yielding a σ value 320 S cm^{-1} (Table S1, Supporting Information). Further, as shown in Figure 1c, EMIM OTF also served as a plasticizer for the composite, with the stretchability increasing from 390% to 630%. It is worth noting that the presence of PAAMPSA did not significantly compromise the σ value compared with the previously reported PEDOT:PSS/soft-polymer systems.[13,21,27] To further explain this, we utilized atomic force microscopy to characterize the morphology evolution with and without ionic liquid. A more pronounced phase separation with fibrillar-like structures was observed in the EMIM OTF-modified PEDOT:PSS film and

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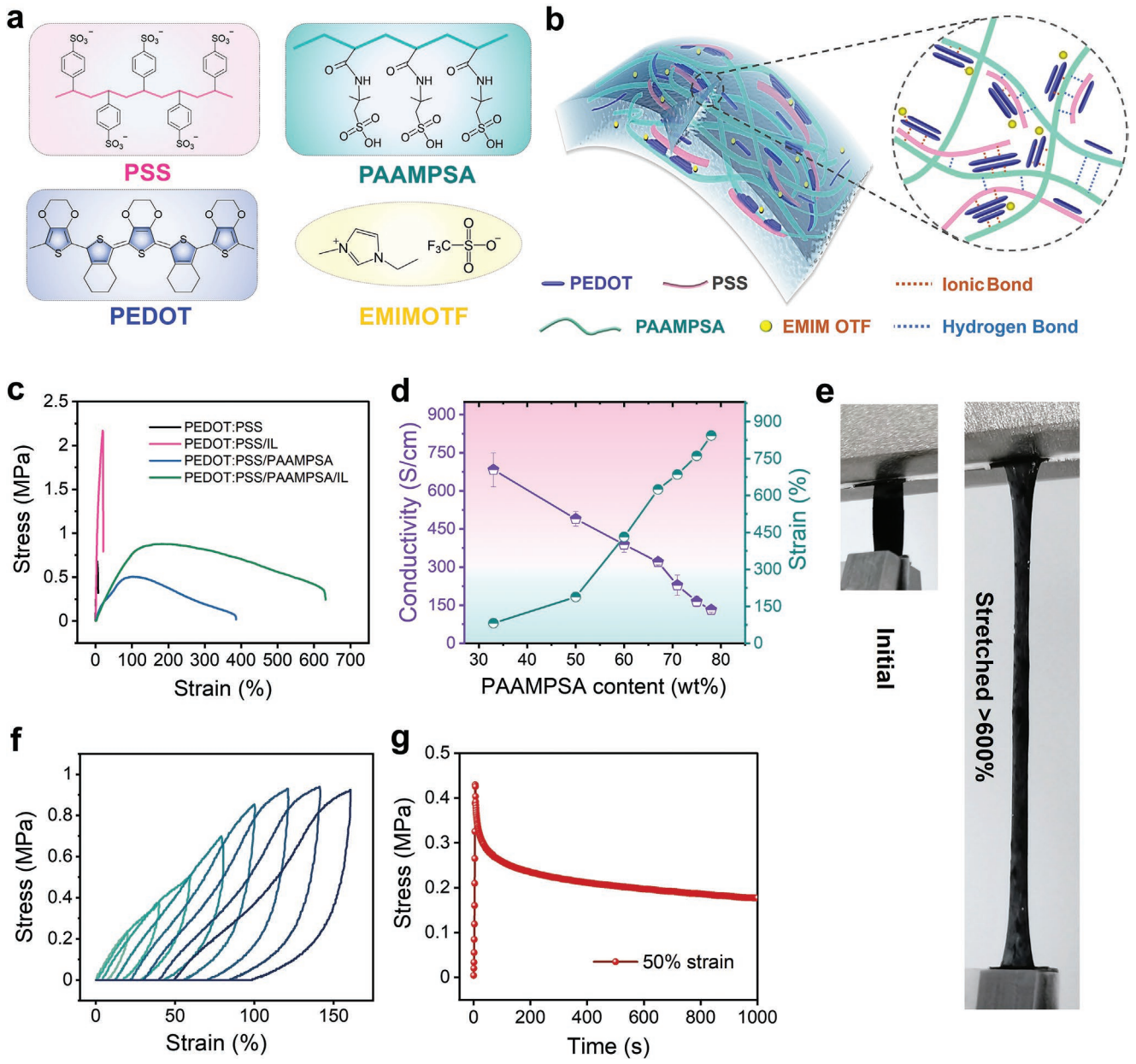


Figure 1. PEDOT:PSS/PAAMPSA/IL films. a) Molecular structures and b) schematic illustration of PEDOT:PSS/PAAMPSA/IL composites. The crosslinking forms regenerative dynamic networks through hydrogen bonds and electrostatic interactions between PEDOT, PSS, PAAMPSA, and IL. c) Stress–strain curves of the hybrid films. d) Conductivity and stretchability of PEDOT:PSS/PAAMPSA/IL with different PAAMPSA ratios. Data presented as mean ± SD, $n = 3$. e) Photograph of a free-standing ternary hybrid film during stretching test. f) Strain cycling and g) stress relaxation curves of PEDOT:PSS/PAAMPSA/IL films.

PEDOT:PSS/PAAMPSA composite, while granular-like structure was observed in the pristine PEDOT:PSS film (Figure S2, Supporting Information). In addition, we observed enhanced PEDOT crystallinity in PEDOT:PSS/IL and PEDOT:PSS/PAAMPSA/IL film as compared with PEDOT:PSS/PAAMPSA film (see the X-ray diffraction spectra in Figure S3, Supporting Information). The fibrillar-like structures and enhanced PEDOT crystallinity in PEDOT:PSS/IL and PEDOT:PSS/PAAMPSA/IL facilitate the charge/hole transport intra- and inter-PEDOT chains. These results indicate that the presence of PAAMPSA does not impede the formation of higher PEDOT crystallinity induced by EMIM OTF, and therefore enabling high electrical conductivity.[26] The slightly lower σ value in PEDOT:PSS/PAAMPSA/IL as compared with PEDOT:PSS/IL film, is mainly attributed to the deceased PEDOT volume fraction in the overall composites.

To elucidate the effect of PAAMPSA on the mechanical property of the nanocomposites, a series of tensile tests using free-standing PEDOT:PSS/PAAMPSA/IL films (≈150 µm) with various PAAMPSA weight fractions were conducted. Figure 1c and Figure S4, Supporting Information, illustrate that the elongation increases by progressive increase in the PAAMPSA

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weight fraction. In particular, a remarkable stretchability of 630% strain was observed in film containing 67 wt% PAAMPSA, which was about two orders of magnitude higher than that of pristine PEDOT:PSS film (≈7%). The stress-strain curves of the films exhibit typical viscoelastic properties[22] of an initial elastic region, followed by yielding, staining softening, and a break point. The stress drop after yielding is related to structural change via the plastic strain, where the deformation is non-reversible. Although higher stretchability was achieved at the expense of its associated electrical conductivity (Figure 1d) (due to the reduced PEDOT fraction), a composite film containing 78 wt% of PAAMPSA offered a remarkable stretchability of over 800% strain while simultaneously presenting a good electrical conductivity (σ) of 130 S cm^{-1} (Figure 1d). Overall, a balance between electrical and mechanical properties existed (Figure 1d). An optimized ratio can be chosen for specific requirements/applications. It is worth noting that the films with 60 and 67 wt% of PAAMPSA simultaneously lead to conductivity above 300 S cm^{-1} and maximum tensile strain over 300%. In particular, the film containing 67 wt% of PAAMPSA leads to excellent extensibility (630%, Figure 1e), which is around 1.5 times higher than the film with 60 wt% of PAAMPSA, while maintaining a comparable electrical conductivity (320 S cm^{-1}). We therefore chose the optimized ratio of 67 wt% of PAAMPSA (with 16.6 wt% PEDOT:PSS and 16.6 wt% IL) for further investigation.

To further characterize the mechanical properties of the PEDOT:PSS/PAAMPSA/IL films, strain cycling, stress relaxation, and rheological tests were carried out. The stress-strain curves showed hysteresis between loading and unloading. When the film was stretched beyond 60% of strain, the film length change became only partially reversible, indicating the viscoelastic response of the film (Figure 1f). A relaxation time of approximately 5.6 min was measured (Figure 1g). Rheological studies (strain sweep and frequency sweep measurements) were conducted as a function of shear strain and frequency, respectively. The storage modulus is higher than the loss modulus in the linear viscoelastic region (Figure S5, Supporting Information), indicating strongly viscoelastic solid-like characteristic of the films (dominance of elastic nature over viscous nature).[1,28] The dynamic hydrogen and ionic bonds formed in PEDOT:PSS/PAAMPSA/IL could easily be broken and recombined to dissipate energy upon strain, thus making the film highly stretchable at ambient conditions. It should be noted that the remarkable flexibility and stretchability of the PEDOT:PSS/PAAMPSA/IL originated from their intrinsic mechanical properties at ambient conditions without using additional stretchable substrates. Our prepared rectangle-shape free-standing PEDOT:PSS/PAAMPSA/IL film maintained well its dimensions and can be easily peeled off from supporting substrate even after 8 months stored at room temperature (Figure S6, Supporting Information).

The mechanical properties of PEDOT:PSS/PAAMPSA/IL are also greatly influenced by relative humidity (RH). We conducted tensile test of PEDOT:PSS/PAAMPSA/IL films at various RH levels. The σ values of PEDOT:PSS/PAAMPSA/IL films were found to be greatly influenced by RH levels. As can be seen in Figure S7 and Table S2, Supporting Information, PEDOT:PSS/PAAMPSA/IL showed a σ of 252.6 S/cm at 30% RH and increased to 363.5 S cm^{-1} at 85% RH. In addition, the maximum strain of the film also showed similar humidity dependency, increasing from 110% to 880%, when the humidity increases from 30% to 85% (see Figure S6, Supporting Information). Such a high humidity dependency of both electrical and mechanical properties is ascribed to the enhanced dynamic electrostatic interaction and hydrogen bonding interaction in PEDOT:PSS/PAAMPSA/IL with moisture adsorption.

To perform the electrical healing experiments, we biased the PEDOT:PSS/PAAMPSA/IL film at a constant voltage of 0.1 V, followed by cutting the film with a razor blade (**Figure 2**a). The damaged PEDOT:PSS/PAAMPSA/IL films showed a fast recovery of electrical conductivity. The film conductivity recovered ≈100% of the original value at ambient conditions upon multiple cuts, with a rapid healing time of $\tau \approx 900$ ms. The healing of electrical conductivity is also visualized by using the PEDOT:PSS/PAAMPSA/IL film as an electrode connected in a circuit with a light-emitting diode (LED) lamp (Figure 2a, and Movie S1, Supporting Information). The LED was switched off after a severe cut using a razor blade and switched on immediately with no significant variations of healing time. Note that the autonomic self-healing time (i.e., recovery time) was heavily dependent on the gap/size of the incision—within 2 h for cut width ranging from 1–5 um and overnight (12–15h) for cut width ranging from 5–15 um (Figures S8 and S9, Supporting Information). We postulated that this healing behavior to be mainly attributed to the highly hygroscopic nature of PAAMPSA in the nanocomposite at ambient conditions (≈23 °C RT, ≈50% RH). Once the polymers were mechanically damaged, the water-uptake capability of the film allowed the swelling of PAAMPSA and PSS. This enabled the polymer chains to diffuse into the damaged area, followed by the reformation of dynamic bonds (like hydrogen bonds and ionic bonds) among PEDOT:PSS and PAAMPSA, which led to the repair of the incision. The water-uptake capability of the film plays a critical role in the self-healing feature at room temperature, while the RH level mainly contributes to the healing efficiency (see Table S2, Supporting Information).

To gain insight into the mechanical strength recovery of PEDOT:PSS/PAAMPSA/IL after cut-and-heal, a freestanding film (≈100 μm) was cut into two pieces by a blade, followed by reconnecting the two separated pieces without any external stimuli. The damaged area healed under ambient conditions (≈23 °C RT, ≈50% RH) without any external stimuli such as external force, light, heat, or adding extraneous solvents (Figure 2b). Tensile tests were performed on the healed free-standing PEDOT:PSS/PAAMPSA/IL films after 24 h at ambient condition (≈23 °C RT, ≈50% RH) (Figure 2b). A similar stress–strain curve was observed for this healed film, indicating recovery of its mechanical properties (Figure 2c). The film exhibited high stretchability after mechanical self-healing, with a maximum strain of 510%, which is the highest reported value so far for autonomic healable PEDOT:PSS-based systems.[13,17,18,21,22,27] The SEM image further verified the physical healing property (Figure 2b). The mechanical self-healing efficiency (determined from the restoration of fracture strain) is higher than >80%, and no significant change in Young's modulus (≈0.8 MPa) was observed before and after healing (Figure 2c).

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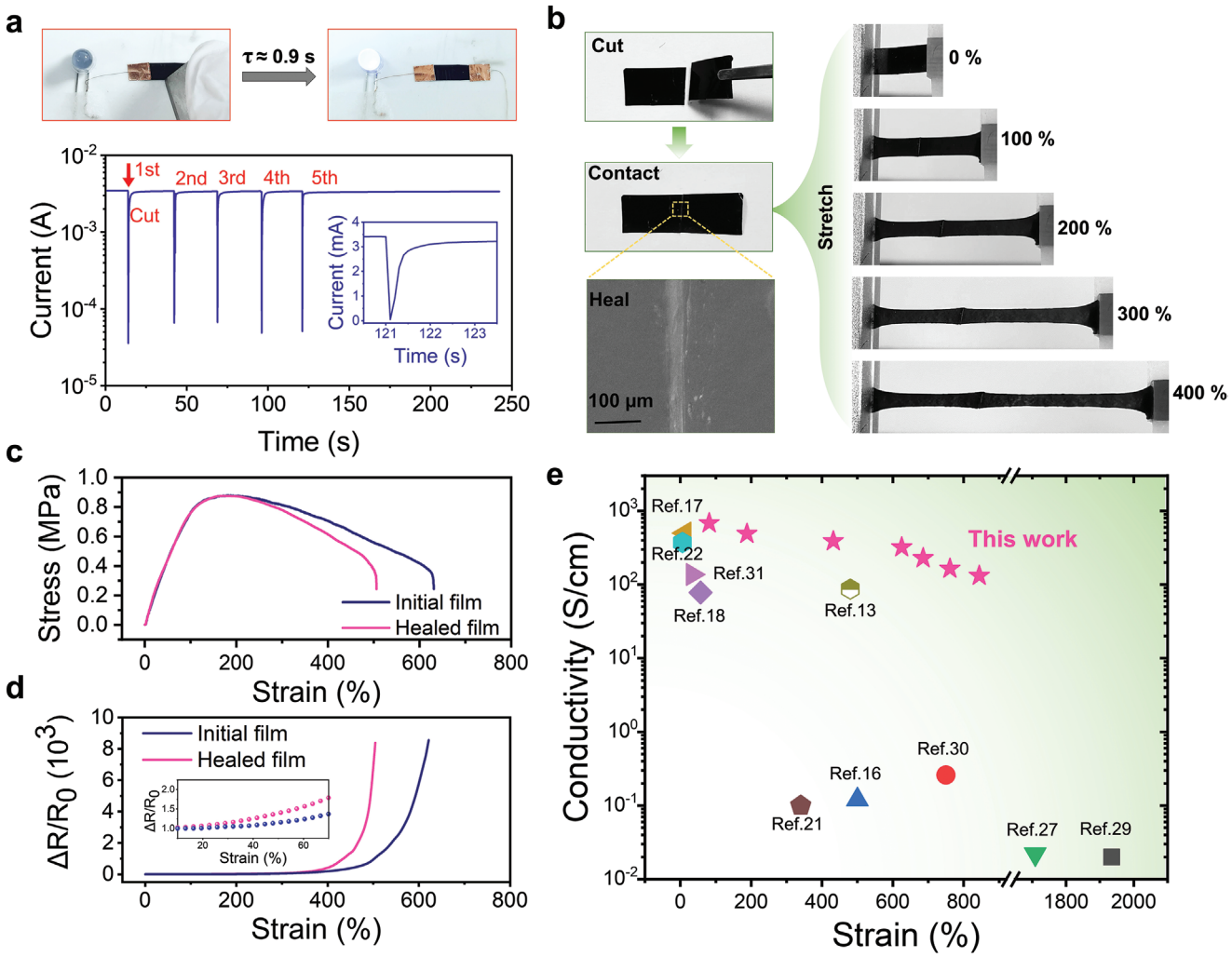


Figure 2. a) Current change of the self-healable PEDOT:PSS/PAAMPSA/IL film with repeated cut-and-heal cycles. b) Photographs of freestanding PEDOT:PSS/PAAMPSA/IL film during consecutive mechanical conditions of cutting apart, recontacting, self-healing, and stretching. c) Stress/strain behavior of free-standing PEDOT:PSS/PAAMPSA/IL film before and after healing. d) The corresponding resistance changes under stretching for both original and self-healed films. Inset is the zoom-in image from 10–70% strain. e) Comparison of conductivity and stretchability presented in this work to representative self-healable and stretchable conductors reported in the literature.

We further recorded the electrical conductivity/resistance changes of original and healed film during the stretching process. A slow linear increase was observed when the stretching was within the elastic deformation region (with the film undergoing 10–70% of strain, Figure 2d). Moreover, under the elastic deformation region, both the original and healed film maintained excellent mechanical and electrical properties, implying the great potential of using this film as an electrode or active material for wearable electronics. When the stretching is beyond the elastic deformation under plastic strain, the resistance increased moderately, which could be attributed to the deterioration of PEDOT morphology, and raised dramatically near the breaking point. This result indicates that the conductive pathways are dynamically reconstructed to recover the mechanical and electrical properties even after serious damage (Figure 2a–d). This "dynamic" healing process is visualized by connecting PEDOT:PSS/PAAMPSA/IL film to a circuit with an LED under consecutive mechanical conditions of stretching, release, cutting, self-healing, stretching after healing, and release (Figure S10, Supporting Information). This dynamic nature of autonomic self-healing PEDOT:PSS/PAAMPSA/IL films with tuneable stretchability and conductivity (Figure 1d) therefore broadens its feasibility for various applications.

Compared with state-of-the-art self-healable and stretchable conductors, our PEDOT:PSS/PAAMPSA/IL films display high electrical conductivity while simultaneously offering great stretchability and excellent autonomic self-healing properties at ambient conditions (Figure 2e and Table S3, Supporting Information).[13,16–18,21,22,27,29–31] Although either higher stretchability or electrical conductivity have been demonstrated in previous reports, none of them exhibit all these three features simultaneously before and after healing. It is worth noting that the negligible drop in conductivity under tensile strain at elastic region is particularly useful for wearable electronics.

The introduction of PAAMPSA which not only affords the desired mechanical properties but also includes self-healing

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feature, while maintaining the electrical conductivity of the composite film can be of great interest in robotics and biomechanical systems applications. In particular, the development of tactile sensor is of critical importance. One of the common strategies to improve the pressure sensitivity of the tactile sensor is to coat an inorganic/metal conducting layer on microstructures (such as pyramid structure).[32,33] However, the mismatched Young's modulus and poor tensile strains (≈2%) have become the major challenges when a high degree of deformation occurs,[33–35] resulting in serious cracking structures on soft elastomeric substrates. Also, the typical metal materials such as Pt or Ag usually show high electrical conductivity but small sensing range (<20 KPa).[32,33] For these reasons, organic conductive materials that offer excellent mechanical properties have been strongly desired.[36] Additionally, due to long-term deformation, a conductive material with robust property and self-healing ability will further extend the device lifetime. However, it is hard to achieve high sensitivity, fast response time, and low hysteresis due to the viscous nature of the self-healing materials. We therefore conducted more detailed studies using our autonomic self-healable and highly stretchable conductors for robust tactile sensors, to demonstrate their potential for wearable electronics. Our tactile sensor consisted of PEDOT:PSS/PAAMPSA/IL composite coated on a microstructured styrene–ethylene–butylene–styrene (SEBS) elastomer film and patterned flexible electrodes, with the device architecture presented in **Figure 3**a. The SEBS film is patterned into a pyramidal array with a uniform pyramid diameter of 10.5 µm, onto which PEDOT:PSS/PAAMPSA/IL is coated using a spray-coating printing technique. High-resolution bottom and cross-section SEM images indicated uniform coating of the PEDOT:PSS/PAAMPSA/IL composite, with a thickness of ≈500 nm, on the SEBS pyramidal structures. Upon the application of an external force, the microstructured sensing layer deforms, leading to an increased contact area between the sensing layer and electrode layer, thereby decreasing its resistance.

The performance of tactile sensor was first characterized by recording the current change under varying pressure, at a

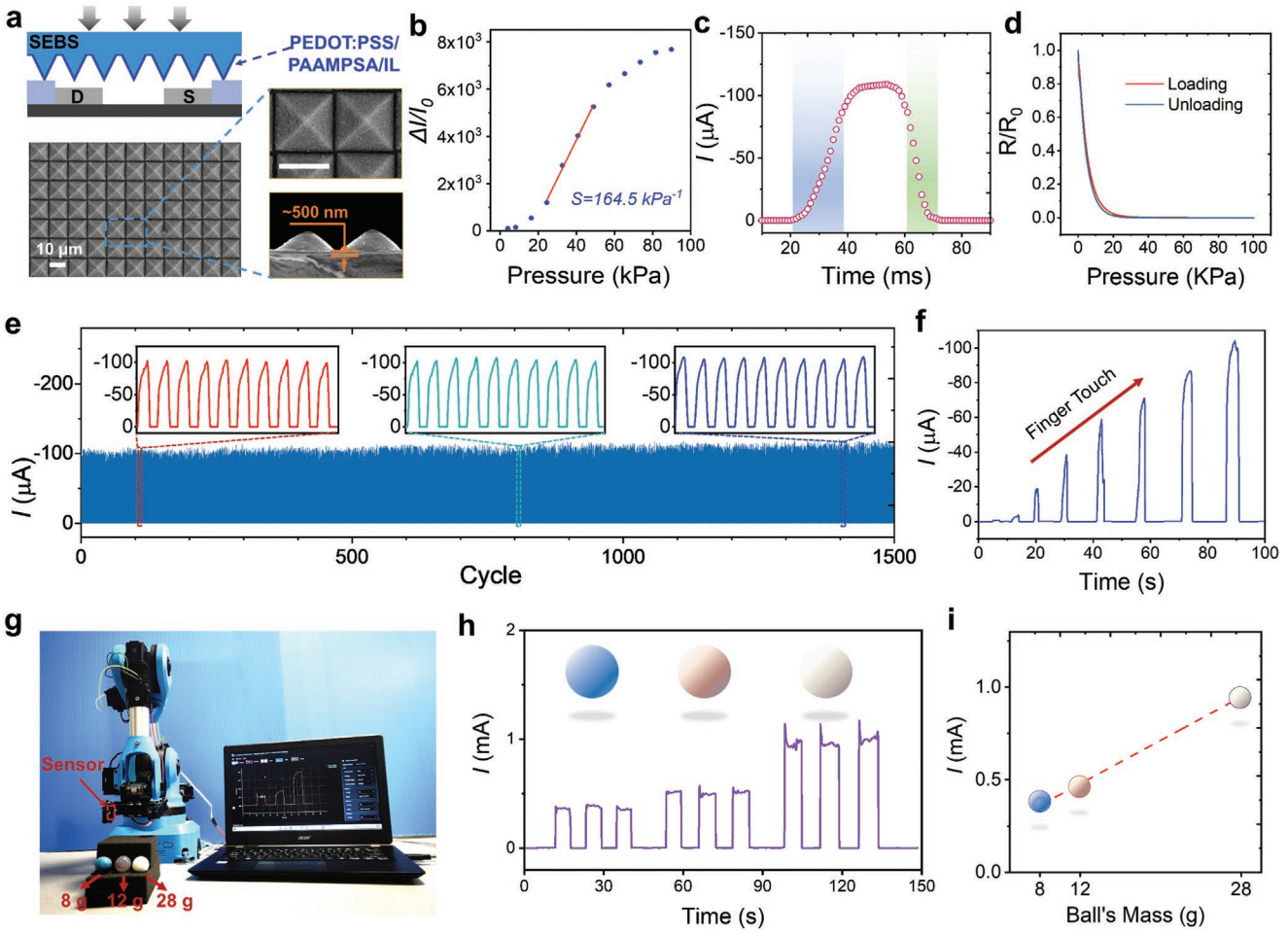


Figure 3. Characterization and sensing performance of tactile sensor using the PEDOT:PSS/PAAMPSA/IL system. a) The schematic of the tactile sensor and SEM images of the micropatterned SEBS substrates coated with PEDOT:PSS/PAAMPSA/IL composite. b) The linear response of pyramidal tactile sensors to applied pressure. c) Response and recovery time of the tactile sensor to an applied pressure of 60 kPa. d) The response of the sensor at pressure range from 0 to 100 kPa in a loading/unloading cycle. e) Stability of the sensor is demonstrated by applying and releasing pressure of 80 kPa for 1500 cycles. f) The current response of tactile sensors in response to finger touch. g–i) The electrical signal monitored by the sensor attached to the robot gripper by grabbing different masses of balls.

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constant voltage of –0.5 V (Figure S11 and Figure 3b, Supporting Information). The pyramidal PEDOT:PSS/PAAMPSA/IL film illustrates a sensitivity of 164.5 kPa^{-1} with a fast response time of 19 ms and recovery time of 13 ms (Figure 3c). The inherently elastic feature of the elastomer and the small shape factor of pyramids allowed a high deformation ability upon compression due to the strain concentration. Additionally, the tactile sensor showed a low hysteresis (DH = 11%, degree of hysteresis, defined as the ratio of area difference between loading and unloading cycle of the resistance–pressure curve, Figure 3d). This low hysteresis can be attributed to the low viscous property of PEDOT:PSS/PAAMPSA/IL and micropyramidal design of tactile sensor with an air gap between the pyramidal film and electrode.[33,37] The excellent stability of the tactile sensor was verified by the negligible performance variation of 4.7% during repeated application and release of pressure for 1500 cycles (Figure 3e). These features are advantageous for a sensitive, reliable, and real-time transduction of pressure signals. Figure 3f presents the current change in response to varying pressure applied using finger touch. As such, the pressure sensor could potentially be used in human-machine interaction systems to monitor robotic components. We further attached the sensor onto a robotic gripper, as illustrated in Figure 3g, Figure 3h, and Movie S2, Supporting Information, demonstrated the real-time recording of grabbing behaviors with various mass of balls. The obtained current signals could be linearly correlated to the mass of the grabbed ball, indicative of the great potential of our tactile sensor in weighing the corresponding object mass (Figure 3i).

Despite progress in developing micro-scale or free-standing self-healable conducting films, there are limited studies on exploring nano-meter thick self-healable conducting polymer for flexible electronics. A thin film of self-healable conducting materials is crucial when used as active layers in developing flexible wearable electronics. We therefore prepared thinner films of PEDOT:PSS/PAAMPSA/IL on SEBS elastic substrates (Figure S12a, Supporting Information) using a spray-coating technique and evaluated their mechanical and electrical properties.[38] The conductivity of the spray-coated thin film (≈200 nm) was found to be higher than thick drop-casted films (≈100 µm) (Figure S12b, Supporting Information). This variation could be probably attributed to the morphology difference due to processing methods and drying kinetics.[39] As presented in Figure S12b, Supporting Information, under stretching, an increase in electrical conductivity from 476.5 to 675.7 S cm^{-1} was observed, possibly due to the re-arrangement of PEDOT-rich domains, forming highly ordered molecular structures along the stretching direction.[39] Further strain of 50% or more consistently reduced the electrical conductivity, reaching a final conductivity of 9.4 S cm^{-1} when stretched over 800% strain, beyond which point the substrate ruptured.

We next subject the spray-coated thin-film to damage by razor blade cuts. We note that the damage gap was unevenly distributed along with the cutting (Figure S12c,d, Supporting Information)—the maximum cut width was larger than 10 µm, while the narrow region was smaller than 1 µm and there are still small connecting paths. It was found that the narrow region of the gap can be mechanically healed within 6 h after cutting, while the largest gap needs 2 days to complete the healing process. The electrical performance was immediately recovered within 30 s (Figure S12e and Movie S3, Supporting Information) after cutting, in line with previous studies on the self-healing behavior of free-standing films as well as possibly due to existence of the small connecting paths in the damaged area. The current measurement shows that the film retained over 92% of its original value after 3 times of cutting. Additionally, we noticed that a gap smaller than 15 µm is needed for autonomic healing of thin films at room conditions. The self-healing feature disappeared when the thin film (200 nm) was cut with a width larger than 15 µm. For larger gaps, external stimuli are needed. Figure S12f shows that a ≈18 µm gap was mechanically healed within 5 s after being exposed to water vapor. This phenomenon further verified the important contribution of the dynamic non-covalent hydrogen bonds for healing as illustrated in Figure 1b.

Finally, we proceed to fabricate fully stretchable organic electrochemical transistors (OECTs) (5 × 4 arrays; **Figure 4**a). The OECT consists of organic channel (PEDOT:PSS/PAAMPSA/IL) that was in direct contact with a solid electrolyte (poly(vinylidene fluoride-hexafluoropropylene)/EMIM OTF). Source/drain metal electrodes based on silver nanowires (AgNWs)) make electrical contact to the channel, while a gate electrode based on single-walled carbon nanotube (SWCNT) makes electrical contact to the electrolyte used to modulate ion migration in or out of the organic channel. The OECT device was configured with a channel length L of 250 µm and a channel width W of 1000 µm, full details on device fabrication can be found in the experimental section. All the transistors were prepared with a comparable channel thickness of ≈200 nm. Figure 4b,c show representative transfer and output characteristics of the PEDOT:PSS/PAAMPSA/IL-based OECTs. Upon application of a positive gate bias, the cations ($EMIM^+$) from the solid electrolyte are injected into the channel, beginning the de-doping process of PEDOT (oxidized state of $PEDOT^{2+}$ and $PEDOT^+$ are reduced to neutral state of $PEDOT^0$), and, as a result, the drain current reduced. These PEDOT:PSS/PAAMPSA/IL OECTs showed high transconductance of 12.95 ± 1.85 mS (average of 20 channels), with the highest value reaching 16.8 mS (Figure 4d and Figure S13, Supporting Information), fast transient response of 7.1 ms (Table S4 and Figure S14a, Supporting Information), and good ON/OFF ratio of $\approx 10^3$ (Figure S14b, Supporting Information). It is worth noting that high-performance uniformity was achieved in this OECT array with narrow on-current and g_m distributions (Figure 4e). The pulse measurement presented in Figure S15, Supporting Information, verified the superior electrochemical stability of PEDOT:PSS/PAAMPSA/IL channel over 1000 continuous operating cycles.

We also explored the OECT performance under mechanical strain both along and perpendicular to the channel length directions. Upon application of strain (from 0% to 100%) parallel to the channel (L direction stretching), we observed a linear decrease in both drain current and g_m, from 5.07 to 1.22 mA and 13.39 to 4.46 mS, respectively (Figure 4f and Figure S16, Supporting Information). Most of the changes in drain current and g_m value correlated well with the change in the channel geometry induced by the strain (Table S5, Supporting Information). Furthermore, the current and g_m were observed to recover 68% (3.44 mA)

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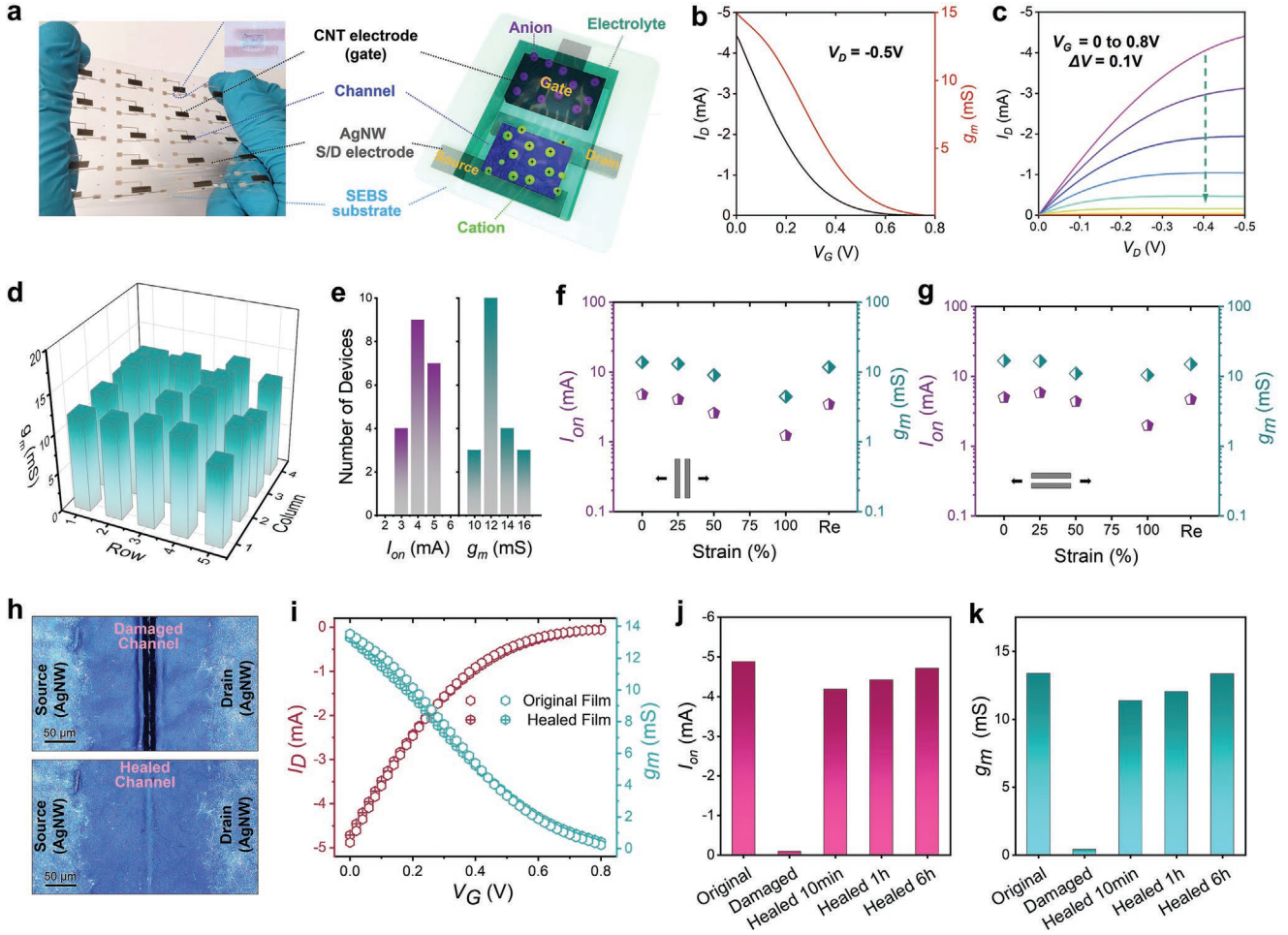


Figure 4. Stretchable and self-healable PEDOT:PSS/PAAMPSA/IL OECTs array. a) Photograph of a 5 × 4 OECT array and schematic illustration of the OECT structure. Inset: Microscope image of one transistor from the array. b) Representative transfer characteristic of a PEDOT:PSS/PAAMPSA/IL -based OECT device. I_D, drain current, represented by black line; g_m, transconductance, represented by the red line; The applied drain-source voltage V_D = –0.5 V. c) The corresponding output curve, with V_G varying from 0 to 0.8 V. d) Transconductance distribution in the 5 × 4 transistor array. e) Statistical distribution histograms of on-current (left) and g_m (right) for transistor array. f,g) Change of on-current and g_m during stretching up to 100% strain along the directions parallel and perpendicular to the channel. h) Optical images of the transistor before and after cutting and healing. i) Transfer curves of an original transistor and healed after damage, at V_D = –0.5 V. j,k) On-Current and g_m of damaged and healed OECTs.

and 88% (11.76 mS) respectively upon releasing the applied strain. The change in drain current and g_m value possibly comes from the change in interface contact resulting from stretching-induced local plastic deformation, and also from the slight delamination between PEDOT:PSS/PAAMPSA/IL and SEBS substrate. On the other hand, upon application of strain (from 0% to 100%) perpendicular to the chancel length (W direction stretching), a slight increase in current from 4.98 to 5.80 mA at 25% of strain was observed (Figure 4g), probably owing to the increased W/L ratio and strain-induced alignment of the conjugated polymer fibers.[39] Upon further increasing the strain to 50% and 100%, the corresponding current and g_m values reduced gradually. After releasing the strain, the current and g_m were observed to recover to close to their initial values. All these results suggest that our high-performance OECTs can maintain normal operation and relatively stable device performances while undergoing a high level of mechanical stretching. We note that for most practical applications, the applied strains are only 20% to 30%,[6,40] which our device is able to completely recover upon stress release.

To verify the self-healing behavior of our OECTs, we recorded the corresponding electrical performance changes before and after cutting/healing (Figure 4h–k). The drain current of the transistors recovered over 85% after 10 min of healing (Figure 4j and Figure S17, Supporting Information). Within 6 h of healing, current was observed to recover to close to its initial values (97%). The same trend was also observed for the g_m (Figure 4k). Additionally, we observed a complete disappearance of the micro gap within the damaged films (Figure 4h), indicating the healed film had returned to its original film morphology. To the best of our knowledge, this is the first demonstration of complete autonomic healing of a thin conductive film (≈200 nm) after mechanical damages with a gap in the micrometer range.

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3. Conclusion

In summary, we successfully demonstrated an autonomic self-healable and highly stretchable conducting polymer with great electrical conductivity, via incorporating dynamic bonds including hydrogen bonds and ionic bonds between the conducting polymer of PEDOT:PSS and a soft-material of PAAMPSA. The hydrogen and ionic bonds formed between PEDOT:PSS and PAAMPSA are able to undergo an energy dissipation mechanism through breakage of bonds when strain is applied or upon severe damage from mechanical stretching or blade cutting, while these dynamic bonds can be reformed to repair the damaged area and thus restore its mechanical property at ambient conditions (≈23 °C RT, ≈50% RH). The optimal PEDOT /PAAMPSA film exhibited stretchability as high as 630% and high electrical conductivity of 320 S cm^{-1}, while having the ability to repair both mechanical and electrical breakdowns at ambient conditions when undergoing severe damage. Furthermore, no significant change in Young's modulus (≈0.8 MPa) was observed before and after healing. Using the optimal PEDOT/PAAMPSA film, we successfully demonstrated a tactile pressure sensor exhibiting high-pressure sensitivity (164.5 kPa^{-1}), ultrafast response time as low as 19 ms, and superior operational stability with negligible performance change after 1500 cycles of tests. The fabricated tactile sensor exhibits great practicality to be used in human-machine interaction systems to identify the weight of grabbed objects. Additionally, a large-area stretchable OECTs array with excellent device performance (with high transconductance of 12.95 ± 1.85 mS, fast transient response speed of 7.1 ms, and good electrochemical doping/dedoping stability) was presented. Notably, this OECTs array, having a thin channel thickness of ≈200 nm, exhibits remarkable self-healability at ambient conditions, which further indicates its utility for e-skin electronics. Our findings provide a facile, low-cost but effective strategy to develop a highly stretchable and autonomic healable conductor towards deformable electronics and future large-area industrial production.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

conducting polymers, organic electrochemical transistors, self-healing ability, stretchable conductive materials, tactile sensors

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